



University of Engineering & Management, Kolkata

End Semester Examination, December 2020

Course: B.Tech(CSE/CSIT/CST) Semester: 3rd

Paper Name: Computer Organization & Architecture

Paper Code: ESC302

Full Marks: 100

Time: 3 hours

Answer all the questions. Each question is of 10 marks.

1. **A.** Explain the structure of a IAS computer. How does Von Neumann can be solved using introduction of a cache memory?

OR

B. What are the differences between Harvard Architecture and Von Neumann Architecture? Explain general register based CPU organization with Example. **5+5**

2. **A.** What do you mean by Base index and PC-relative addressing mode? Explain it with the help of an example.

OR

B. Draw and explain the working of a 4 bit ALU circuit with proper functions included.

3. **A.** What are the advantages and disadvantages of serial adder and parallel adder?

OR

B. Explain the difference between direct mapping and set-associative mapping techniques. What are “write-through” & “write-back” cache policies?

4. **A.** What are differences between dynamic pipeline and static pipeline? State the differences between scalar and vector pipeline.

OR

B. Suppose a program (or a program task) takes 1 billion instructions to execute on a processor running at 2 GHz. Suppose also that 50% of the instructions execute in 3 clock cycles, 30% execute in 4 clock cycles, and 20% execute in 5 clock cycles. What is the execution time for the program or task?

5. **A.** What is the speedup that can be obtained on 100 processors if 93% of the program is ideally parallel, the remaining part is sequential? In multiprocessor environment how the speedup increases with the count of processor? Explain with graph.

OR

B. Write a program to evaluate arithmetic expression:

$$X = (A-B) * (C-D)$$

- i)** Using general register computers with 3-address instruction.
- ii)** Using general register computer with two address instruction.
- iii)** Using stack organized computer with zero address instruction

6. A. Design a 4 bit 14 function ALU unit. Design a logic unit to perform AND, OR, NOT, XOR operations.

OR

- B. Draw the circuit of a BCD adder and explain its operations.

7. A. Explain circular shift and arithmetic shift with proper example.

OR

- B. Write short note on the following: (Any two)

2*5

i) Cache memory

ii) Paging

iii) Segmentation

8. A. With proper diagrams, explain Flynn's classification for modern digital computers.

OR

- B. What do you mean by RAM expansion? What are its possible types? Define each with a suitable example.

9. A. i) What is a multiprocessor system? How does it differ from a multicomputer system?

ii) Discuss tightly coupled versus loosely coupled system.

OR

- B. What are the differences between vector and scalar processor?

10. A. Differentiate between polled IO, Interrupt initiated IO and DMA.

OR

- B. Draw and explain the architecture of a hardwired control unit.
